

NFC Card and Tag Security Enhancement Framework Using Hybrid Detection

NFC Security Profile Scoring Model

J26-CS-316

Project Proposal Report

Anjana I. K .D. - IT23160620

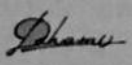
B.Sc (Hons) Degree in Information Technology Specialized in
Cyber Security

Department of Information Technology
Sri Lanka Institute of Information Technology Sri Lanka

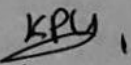
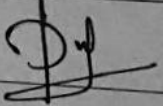
September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Anjana I. K. D.	IT23160620	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
K. P. Yapa	15/09/2026	
Ms. Divyathi Pandithage	15/09/2026	

ABSTRACT

Access control systems, university ID systems, transportation cards, hotel access cards, and smart information sharing applications all make extensive use of Near Field Communication (NFC) cards and tags. Important security-related features on these cards include memory structures, protection settings, NDEF data, authentication methods, and unique IDs. However, the majority of NFC tools currently in use primarily explain a card's technical characteristics, such as its family, UID length, memory structure, and NDEF content. They don't offer a consistent, universal security score to gage an NFC card or tag's true exposure level. This study suggests an Exploitability-Calibrated NFC Security Profile Scoring Model to overcome this constraint. Measurable security parameters like UID writability, authentication type, default key presence, memory lock state, NDEF protection, and known vulnerability indications are used by the suggested approach to assess NFC cards and tags. Authorized NFC test cards are tested in a lab setting after being programmed into various security settings. T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 are the three attacker capability categories identified by the model. The experiment logs read success, write success, emulation possibility, number of attempts, compromised state, and time-to-compromise for every card configuration and attacker tier. To create calibrated score weights, the gathered data is analyzed using Cox regression and stored in structured CSV datasets. Ultimately, each card is categorized as Low, Medium, or High risk based on the model's 0–100 NFC security risk score. This makes evaluating NFC card security more transparent, data-driven, and attacker-tiered.

Keywords: NFC Security, RFID, Security Scoring, Time-to-Compromise, Cox Regression, Flipper Zero, Proxmark3, Attacker-Tiered Model, Risk Classification

ACKNOWLEDGEMENT

I sincerely thank my research supervisor and co-supervisor for their valuable guidance, continuous support, and constructive feedback throughout the development of this research proposal. Their academic and technical advice helped me define the research scope, methodology, and expected contribution of this component.

I would also like to thank the Department of Computer Systems Engineering, Faculty of Computing, and the Sri Lanka Institute of Information Technology for providing the academic environment, resources, and facilities needed to carry out this final-year research project.

I also extend my appreciation to the relevant academic and research committees for their guidance on the research direction, ethical considerations, and the authorized use of specialized NFC/RFID equipment within a controlled laboratory environment.

I want to thank the members of my research group for their cooperation, discussions, and exchange of ideas while working on the related components of the NFC/RFID Card and Tag Security Enhancement Framework. Their collaboration has contributed to the overall development of the research while maintaining the independence of each component.

Finally, I would like to express my heartfelt gratitude to my family and friends for their patience, encouragement, and continued support throughout the research planning process.

TABLE OF CONTENTS

DECLARATION.....	2
ABSTRACT.....	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENTS	5
LIST OF FIGURES	7
LIST OF TABLES	8
LIST OF ABBREVIATIONS	9
1. INTRODUCTION	10
1.1 Background and Context.....	10
1.2 Literature Review.....	13
1.3 Research Gap	14
2. OBJECTIVE	16
2.1 Main Objective.....	16
2.2 Specific Objectives	16
3. METHODOLOGY	17
3.1 Research Methodology and Design	17
3.2 Data, Participants and Experimental Inputs.....	18
3.3 Data Preparation and Dataset Design	20
3.4 Cox Regression-Based Weight Calibration.....	21
3.5 Risk Score Calculation and Classification.....	22
3.6 Validation Strategy and Performance Metrics	23
3.7 Tools and Platforms with Justification.....	25
3.8 Ethical Considerations and Data Collection Limitations.....	26
4. HIGH-LEVEL SYSTEM ARCHITECTURE.....	27
4.1 Overview of the Proposed Solution	27
4.2 Internal Architecture of the Exploitability-Calibrated NFC Security Profile Scoring Model	28
4.3 Data Flow of the System.....	29
4.4 Scalability Considerations	30
4.5 Security Considerations	31
5. USER REQUIREMENTS	32

5.1 Functional Requirements	32
5.2 Non-Functional Requirements	33
6. COMMERCIALIZATION PLAN	35
7. BUDGET AND JUSTIFICATION	36
8. WORK BREAKDOWN STRUCTURE (WBS).....	37
9. GANTT CHART	39
9.1 Project Timeline	39
9.2 Risk Register and Contingency Plan.....	40
10. REFERENCES	41
11. APPENDICES.....	43
Appendix A - Dataset Schema and Field Definitions	43
Appendix B - Validation and Evaluation Checklist	44
Appendix C - Ethical Testing Boundary	44
Appendix D — Prototype Dashboard	45
Appendix 2 - Mandatory Proposal Evidence and Decision Log.....	45
Appendix 3. AI Use Disclosure	46

LIST OF FIGURES

Figure 1: Circular Validation in the Existing Weighting Approach.	12
Figure 2: Proposed Research Pipeline from Measurement to Calibrated Score	18
Figure 3: Illustrative Tier Matrix	23
Figure 4: High-level architecture of the four-component framework.....	27
Figure 5: Internal Architecture of the Exploitability-Calibrated NFC Security Profile Scoring Model	29
Figure 6: Python-Based NFC Security Scoring Tool Workflow.....	30
Figure 7: Work Breakdown Structure	38
Figure 8: Gantt Chart	40
Figure 9 Prototype Dashboard	45

LIST OF TABLES

Table 1: Comparison of Existing NFC Assessment Tools and Frameworks.....	12
Table 2:Summary of Reviewed Literature.	14
Table 3:Summary of Identified Research Gaps.	15
Table 4:Card Configuration Matrix for Security States C0 - C7	19
Table 5: Attacker Capability Tiers.	19
Table 6:Per-Trial Measurements Recorded During Testing.....	20
Table 7:NFC Security Score Factors and Coding.	21
Table 8:Dataset Structure for configs.csv and attacks.csv	21
Table 9:Evaluation Metrics and Target Outcomes	24
Table 10:summarizes the Validation Tests for the Proposed Scoring Model.....	24
Table 11: Tools and Technologies Used in the Scoring Model.....	25
Table 12: Functional Requirements of the Proposed Scoring Tool.....	33
Table 13:Non-Functional Requirements of the Proposed Scoring Tool	34
Table 14:Budget and justification.	36
Table 15: Work Breakdown Structure.....	38
Table 16: Risk Register and Contingency Plan.....	40

LIST OF ABBREVIATIONS

3DES – Triple Data Encryption Standard

AES – Advanced Encryption Standard

API – Application Programming Interface

C1 – Component 1 – NFC Security Profile Scoring

C2 – Component 2 – Multi-Session Protocol-Response Provenance Fingerprinting

C3 – Component 3 – NFC Tag Data Integrity Verification

C4 – Component 4 – Hybrid Behavioral Anomaly Detection

CSV – Comma-Separated Values

CVSS – Common Vulnerability Scoring System

FR – Functional Requirement

HF – High Frequency

IEEE – Institute of Electrical and Electronics Engineers

MITM – Man-in-the-Middle

NDEF – NFC Data Exchange Format

NFC – Near Field Communication

NFR – Non-Functional Requirement

OTP – One-Time Programmable

RFID – Radio Frequency Identification

T1 – Tier 1 – Smartphone Attacker Capability

T2 – Tier 2 – Flipper Zero Attacker Capability

T3 – Tier 3 – Proxmark3 Attacker Capability

UID – Unique Identifier

WBS – Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

The short-range wireless communication technology known as Near Field Communication (NFC) is extensively utilized in contemporary settings for smart cards, access control, and identification. NFC cards and tags are frequently utilized in digital information exchange applications, hotel key cards, transportation ticketing systems, staff access control systems, university student identity systems, and product identification tags. A Unique Identifier (UID), memory blocks or pages, NFC Data Exchange Format (NDEF) records, authentication settings, and card-specific protection methods are among the security-related data included on these cards, which typically run at 13.56 MHz.

Although NFC cards are extensively used and handy, administrators, auditors, and users may not always be able to see their true security posture. Even if two NFC cards may be issued by the same company and have the same physical appearance, their internal security standards may differ greatly. While one card might have strong authentication and protected memory, another might be open, writable, set up with default keys, or vulnerable due to known flaws. Therefore, determining whether an NFC card or tag is secure requires more than just looking at the card type, UID, or visible NDEF content.

In many real-world scenarios, NFC systems still rely on data from cards that is visible or easily readable. This creates security concerns because a card may seem fine during a basic check but still have insufficient protection. For example, some cards may allow illegal reading, writing, or emulation, depending on how they are set and the tools available to the attacker. Because of this, it may be difficult for facility managers and security auditors to determine the level of risk involved in a present NFC card deployment.

NFC cards can be identified or analyzed using current NFC technologies like Flipper Zero, Proxmark3, and smartphone NFC apps. Card family, UID length, memory structure, NDEF data, and occasionally activity relating to authentication can all be shown by these tools. Nevertheless, the majority of the output produced by these technologies is descriptive. They are able to describe the card, but they can not offer a standardized numerical score that indicates the card's level of danger or exposure.

Organizational risk matrices and CVSS are examples of generic risk assessment frameworks that offer structured risk rating techniques for information systems and software vulnerabilities. These frameworks aren't made especially for contactless physical credentials, though. UID writability, authentication type, default key existence, memory lock state, NDEF protection, and known

vulnerability indications are some of the variables that affect NFC card exposure. There is no clear mapping between these NFC-specific variables and general software vulnerability score metrics.

Therefore, as Novelty Exploitability-Calibrated NFC Security Profile Scoring Model of the NFC/RFID Card and Tag Security Enhancement Framework, this study suggests an NFC Security Profile Scoring Model. The suggested component seeks to categorize NFC cards as Low, Medium, or High risk by converting measurable security features into a 0–100 risk level. The model is intended to be repeatable, explainable, attacker-tiered, and exploitability-calibrated.

The primary difference in this suggested model is that the scoring weights are chosen solely by expert opinion rather than being manually estimated. Rather, approved NFC test cards are set up in various security states and examined in a lab setting. Time-to-compromise is a measure of how long it takes to compromise each configuration. Factor weights are then obtained from this measured laboratory data using Cox proportional hazards regression.

Attacker-tiered scoring is another feature of the suggested paradigm. The model offers distinct scores for three attacker capability levels: T1 smartphone, T2 Flipper Zero, and T3 Proxmark3, rather than providing a single overall score for a card. This is crucial because, depending on the attacker's tools, the same NFC card may carry varying degrees of risk. When examined with Flipper Zero or Proxmark3, a card that seems low risk against a smartphone could turn into medium or high risk.

By offering a configuration-exploitability layer, this component adds to the larger NFC/RFID Card and Tag Security Enhancement Framework. This component's output is a Python-based scoring tool that takes NFC card scan and configuration data, computes a risk score between 0 and 100, and classifies each attacker tier as Low, Medium, or High. The model encourages security administrators, auditors, and researchers by offering an NFC card security posture evaluation technique that is more quantifiable and reproducible.

The limitations of the current NFC assessment methods are shown in Figure 1. Proxmark3, Flipper Zero, and smartphone apps can show testing capabilities or provide card information, but they don't offer a typical numerical security score. By translating NFC card security parameters into a 0–100 attacker-tiered risk score and Low, Medium, or High classification, the proposed Exploitability-Calibrated NFC Security Profile Scoring Model overcomes this restriction.

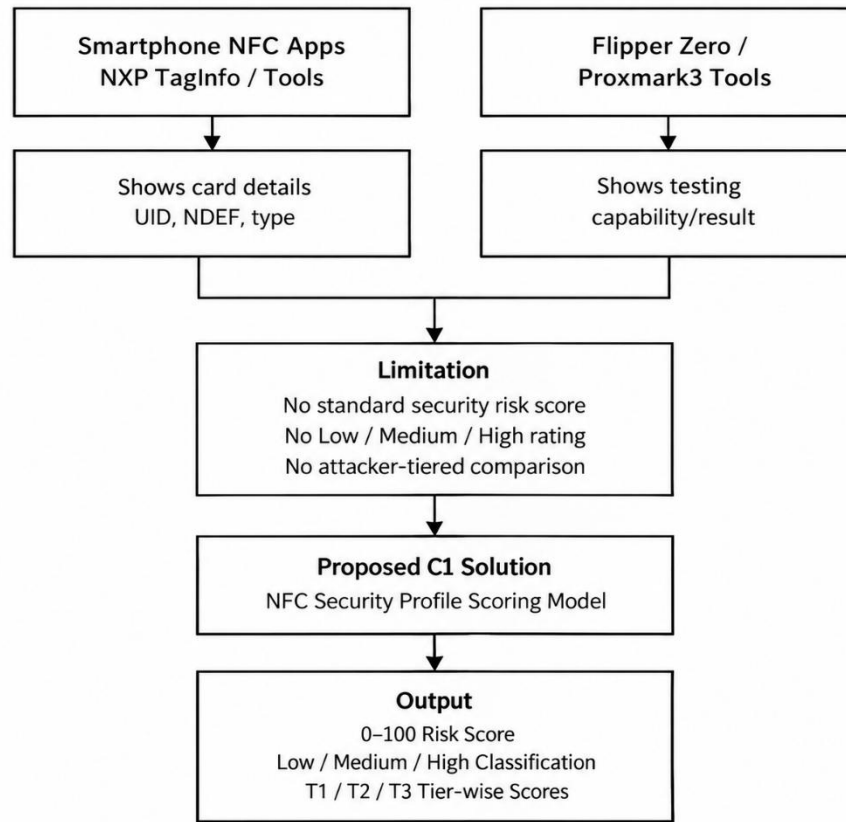


Figure 1: Circular Validation in the Existing Weighting Approach.

Tool / Framework	Main Function	Limitation Addressed by This Research
Smartphone NFC Apps	Read UID, NDEF data, and basic card information	Descriptive only; no security score
Flipper Zero	Read, save, emulate, and test supported cards in authorized lab conditions	Demonstrates capability but does not provide calibrated risk score
Proxmark3	Advanced NFC/RFID protocol-level analysis	Requires expert use and does not automatically produce Low/Medium/High risk rating
CVSS	Generic vulnerability scoring framework	Designed for software vulnerabilities, not NFC card configurations
Expert Rating	Human-based risk judgement	Can be subjective and may cause circular validation

Table 1: Comparison of Existing NFC Assessment Tools and Frameworks.

1.2 Literature Review

1.2.1 Review scope and search method

Access control, identification, attendance monitoring, smart cards, healthcare, and smart tag applications all make extensive use of NFC and RFID technologies. While RFID is utilized for tracking and identification in many settings, NFC is frequently employed for short-range communication between cards/tags and reader devices [4], [6]. These technologies are widely used in real-world systems because they are inexpensive, quick, and simple to use.

Nonetheless, prior studies indicate that NFC/RFID systems may have security flaws. Because weak/default keys and incorrectly configured access bits might enable cloning and tampering in real systems, MIFARE Classic is a frequently cited example [1]. assaults like cloning, skimming, replay, man-in-the-middle, cryptanalysis, and brute-force assaults are also highlighted in a systematic evaluation of RFID-based access control systems [2].

Using programs like Proxmark3, some research focuses on practical security analysis and RFID vulnerability testing [3]. In order to lower the risks of theft, unauthorized duplication, and data tampering, other research suggest security enhancements including fingerprint authentication for RFID/NFC devices [5]. These studies demonstrate that before using cards in sensitive environments, NFC/RFID security needs to be carefully evaluated.

Card details, including UID, card type, memory information, and NDEF data, can be recognized by current NFC/RFID devices. These tools, however, mostly offer technical descriptions. The detected security indicators are not transformed into a distinct 0–100 risk score by them. As a result, users might be aware of the type of card, but they might not be aware of its level of danger.

The significance of controlled testing, structured datasets, and experimental evaluation is further demonstrated by recent RFID investigations. While robustness research employs controlled tests to enhance RFID system performance [9], RFID observation research uses collected data for analysis and validation [8]. The necessity for a quantifiable and reproducible laboratory-based approach for NFC/RFID security assessment is supported by these investigations.

Therefore, laboratory-measured time-to-compromise data and Cox regression-based weight calibration are used in the proposed Exploitability-Calibrated NFC Security Profile Scoring Model. This aids in producing a 0–100 NFC card security risk score independently for several attacker capability tiers, including Proxmark3, Flipper Zero, and smartphones. Table 2 presents the summary of reviewed literature used to support this research.

Reference	Main Focus	Relevance to This Research
[1]	MIFARE Classic cloning and tampering	Supports NFC card security weakness and configuration-based risk
[2]	RFID/NFC access-control vulnerabilities	Supports the research problem and attack categories
[3]	RFID vulnerability and Proxmark3 usage	Supports attacker capability and practical RFID testing
[4]	Smartphone-enabled smart card attendance system	Supports smartphone NFC usage in real-world systems
[5]	RFID/NFC device with fingerprint authentication	Supports NFC/RFID security enhancement need
[6]	RFID and NFC technology background	Supports NFC/RFID operation and application background
[8]	RFID observation dataset processing	Supports structured dataset and experimental analysis idea
[9]	RFID robustness and controlled evaluation	Supports laboratory-based validation approach

Table 2: Summary of Reviewed Literature.

1.3 Research Gap

Despite the widespread usage of NFC/RFID cards in smart card, access control, identification, and attendance systems, current research indicates that these systems still have security flaws [2]. Because cloning and tampering can happen when weak/default keys and incorrectly configured access bits are used, MIFARE Classic is a good illustration [1].

The primary research gap is that the majority of the information provided by current NFC/RFID instruments is descriptive. They are able to display card information, including UID, card type, memory data, and NDEF records, but they can not translate that information into a clear security risk score between 0 and 100.

The fact that standard security scoring methods are not made especially for NFC card configurations is the second research gap. UID writability, authentication type, default key existence, memory lock state, NDEF protection, and card-family vulnerabilities are some of the variables that affect NFC card risk. These elements differ from those that are typically used to score software vulnerabilities.

The absence of attacker-tiered scoring is the third research gap. Smart card systems frequently use smartphone NFC [4], and RFID vulnerability research uses Proxmark3 for more thorough testing [3]. This demonstrates how an attacker's ability can alter the real risk level. For the purpose of evaluating NFC card security, a single general score is therefore insufficient.

The absence of independently measured time-to-compromise data represents the fourth research gap. Current NFC/RFID research facilitates controlled experimental evaluation and organized data collecting [8], [9]. Nevertheless, laboratory-measured compromise time is not used as the ground truth for scoring in current NFC assessment methods.

Consequently, the following is a summary of the research gap:

There is currently no NFC card security scoring methodology that offers distinct results for attacker capability tiers like smartphone, Flipper Zero, and Proxmark3 and generates a repeatable 0–100 risk score using laboratory-measured time-to-compromise data.

The primary research gaps found in the literature are shown in Table 3, along with the suggested contribution of this study.

Identified Gap	Evidence from Literature	Proposed Contribution
NFC/RFID security weaknesses	Cloning, tampering, skimming, replay, MITM, cryptanalysis, and brute-force risks are discussed in NFC/RFID systems [1], [2]	Focuses on measurable NFC card security exposure
Descriptive NFC tools	Existing tools can show card details but do not provide a calibrated risk score	Converts NFC security factors into a 0–100 score
No attacker-tiered scoring	Smartphone NFC systems and Proxmark3-based RFID testing show different capability levels [3], [4]	Gives separate scores for smartphone, Flipper Zero, and Proxmark3
No measured ground truth	RFID studies use structured datasets and controlled evaluation methods [8], [9]	Uses laboratory-measured time-to-compromise as scoring ground truth
Need for security enhancement	RFID/NFC security improvement is discussed through authentication-based protection methods [5]	Provides a defensive scoring model to support security improvement decisions

Table 3: Summary of Identified Research Gaps.

2. OBJECTIVE

2.1 Main Objective

This research section's primary objective is to create a security scoring methodology for NFC cards and tags that will be calibrated using the vulnerabilities' time-to-compromise metrics rather than professional judgment. This will assign a 0 - 100 grade of Low, Medium, or High risk to each of the three attacker capabilities - smartphones, Flipper Zero, and Proxmark3.

2.2 Specific Objectives

1. To determine the primary security features of NFC cards and tags, including known vulnerability indications, memory lock status, default key existence, UID writability, authentication type, and NDEF protection.
2. To create an attacker-tiered testing paradigm utilizing the T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 capability levels.
3. To record read success, write success, emulation success, compromised status, and time-to-compromise in order to gather laboratory-based experimental data from approved NFC test cards and tags.
4. To determine empirical factor weights from measured time-to-compromise data using Cox proportional hazards regression.
5. To create a prototype in Python that determines an NFC security risk score between 0 and 100 and categorizes the outcome as Low Risk, Medium Risk, or High Risk.
6. To verify the suggested scoring model using repeatability testing, baseline comparison, Spearman correlation, and sensitivity analysis

3. METHODOLOGY

3.1 Research Methodology and Design

A quantitative controlled laboratory experimental technique is used in the suggested Exploitability-Calibrated NFC Security Profile Scoring Model. The research attempts to compute a numerical NFC security risk score utilizing quantifiable experimental data rather than subjective expert judgment, which is why this methodology was chosen.

NFC card configuration and attacker capability tier are the two primary experimental parameters that form the basis of the research design. Certain NFC card families, including MIFARE Ultralight, MIFARE Classic, MIFARE DESFire, and NTAG, are set up in various security states. The T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 attacker capability levels are then used to test these combinations.

Measurable variables including read success, write success, emulation success, compromised status, time-to-compromise, censored status, and number of attempts are recorded during each lab session. A card configuration is considered censored data if it is not compromised during the specified 1800-second cut-off period. This enables the model to analyze trial outcomes that are both compromised and uncompromised.

Following data collection, the measured time-to-compromise data is subjected to Cox proportional hazards regression. The scoring model's empirical factor weights are obtained using this technique. Each NFC card or tag is then categorized as Low Risk, Medium Risk, or High Risk for each attacker tier using the calculated weights to compute a 0–100 risk score.

Only the study team's authorized NFC test cards and tags are used in all trials. There will be no testing of actual student cards, staff cards, bank cards, transport cards, hotel key cards, workplace access cards, or third-party credentials. This guarantees that the approach will continue to be morally sound, compliant with the law, and appropriate for defensive cybersecurity research.

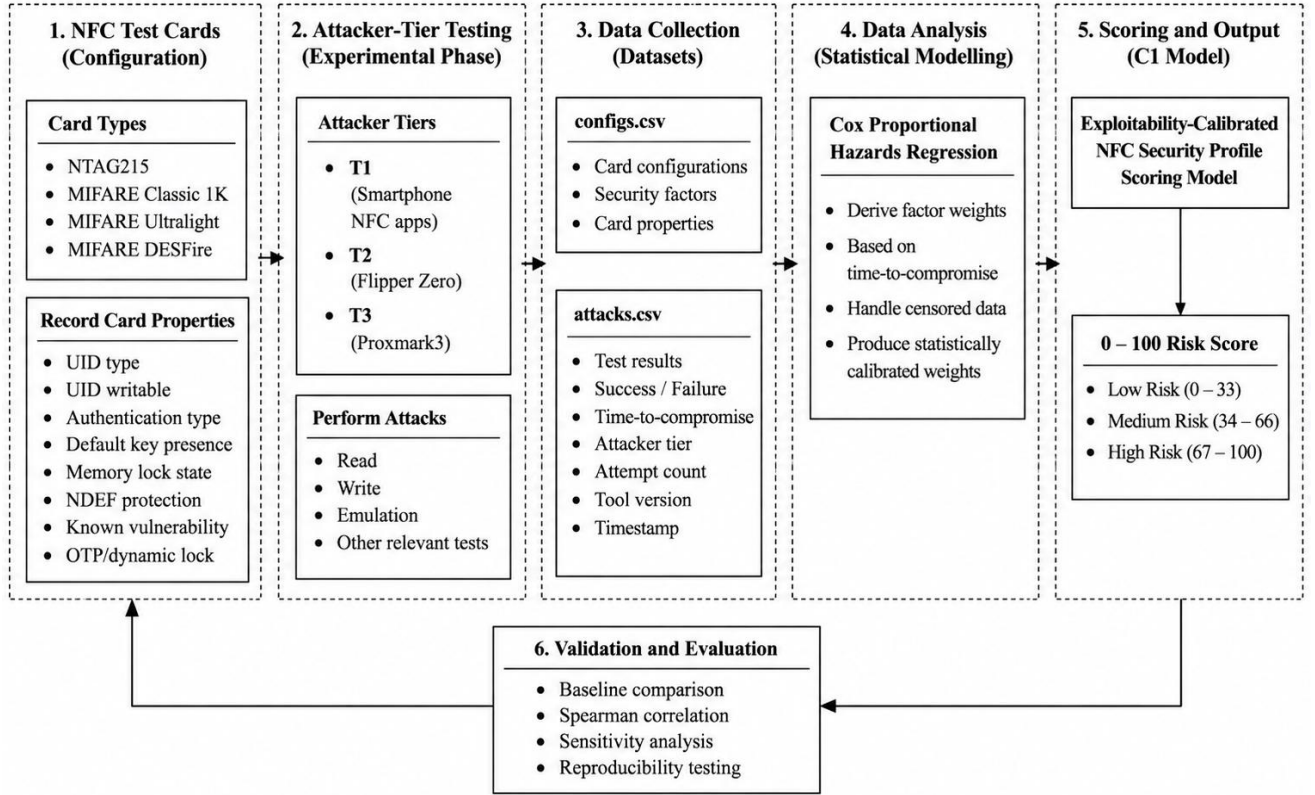


Figure 2: Proposed Research Pipeline from Measurement to Calibrated Score

3.2 Data, Participants and Experimental Inputs

The experimental data gathered from approved NFC test cards and tags is used in the proposed Exploitability-Calibrated NFC Security Profile Scoring Model. Because the necessary data must be gathered through controlled laboratory testing, the study does not rely on a publicly available dataset. To see how each configuration impacts exploitability, several NFC card families, including NTAG, MIFARE Ultralight, MIFARE Classic, and MIFARE DESFire, are set up in various security states.

The suggested NFC card configuration matrix utilized in the controlled lab experiment is shown in Table 3. Every state denotes a distinct security circumstance that may impact an NFC card or tag's exploitability.

State	Description
C0	Factory default / unlocked configuration
C1	Default key changed to a non-default value
C2	Static lock bytes enabled
C3	Dynamic lock or OTP enabled
C4	Password authentication enabled
C5	NDEF set to read-only
C6	Fully locked / maximum available protection
C7	Writable UID / magic card state, where available

Table 4: Card Configuration Matrix for Security States C0 - C7

The attacker capability tier model is the second experimental input. Depending on the attacker's tool, the same NFC card may have varying degrees of exposure. Thus, three attacker capability tiers T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 are defined in this study. The model is able to produce distinct risk rankings for various attacker levels thanks to these tiers.

Tier	Device / Tool	Capability Level
T1	Smartphone	Basic NFC reading, UID inspection, and simple writing on writable tags
T2	Flipper Zero	Practical field-level NFC testing, reading, supported emulation checks, and validation
T3	Proxmark3	Advanced NFC/RFID protocol-level analysis and deeper security testing

Table 5: Attacker Capability Tiers.

One NFC card configuration evaluated under one attacker tier is regarded as one trial for data collection purposes. The experiment documents if reading, writing, or supported emulation-like activity are feasible during each trial. Additionally, it logs the number of attempts, tool version, testing notes, compromised state, time-to-compromise, and censored status. The trial is noted as suppressed data if a card is not compromised within the allotted 1800 seconds.

Measurement	Description
read_success	Records whether unauthorised reading was possible
write_success	Records whether unauthorised writing was possible
emulation_success	Records whether supported emulation or clone-like behaviour was possible
compromised	Records whether any successful compromise occurred
time_to_compromise	Time taken to reach compromise, measured in seconds
censored	Records whether the trial reached the cut-off without compromise
attempts	Number of attempts made during the trial
Tool version	Tool, firmware, or application version used
notes	Special observations for reproducibility

Table 6: Per-Trial Measurements Recorded During Testing.

3.3 Data Preparation and Dataset Design

To compute an empirically calibrated NFC security score, the suggested Exploitability-Calibrated NFC Security Profile Scoring Model needs organized experimental data. The dataset for this study will be produced through controlled laboratory testing because there isn't currently a public dataset that includes NFC card configurations, attacker-tiered trial results, and time-to-compromise figures.

The gathered data will be examined for accuracy, consistency, and completeness before the scoring model is applied. Duplicate records, incorrect labels, missing values, and invalid entries will all be fixed or recorded. Attacker tiers, for instance, need to be consistently reported as T1, T2, and T3. It is necessary to store Boolean values like read success, write success, emulation success, compromised, and censored as either 0 or 1. It is necessary to record the time-to-compromise in seconds.

The suggested scoring approach measures each card or tag configuration's exposure level using NFC-specific security criteria. UID writability, authentication type, default key existence, memory lock status, NDEF protection, and known vulnerability indications are some of these aspects. Each component is transformed into a numerical risk value prior to computing the final score. The codes for safe, medium-risk, and risky conditions are 0, 0.5, and 1, respectively.

Factor	Safe Value	Medium Value	Risky Value
uid_writable	Fixed UID	Not applicable	Writable UID
auth_type	AES / 3DES	Password	No authentication
default_key_present	Changed key	Not applicable	Default key present
lock_state	Fully locked	Partially locked	Unlocked
ndef_protection	Read-only	Not applicable	Read / write
cve_flag	No known issue	Not applicable	Known vulnerability

Table 7: NFC Security Score Factors and Coding.

Two structured CSV files are used to store the experimental data. The laboratory testing results for each attacker tier are stored in the attacks.csv file, whereas the NFC card or tag configuration information is stored in the configs.csv file. The config_id field is used to link the two datasets. This structure makes it possible to combine the measured time-to-compromise findings and security factor values for Cox regression analysis and the computation of the final risk score.

Dataset	Purpose	Main Fields
configs.csv	Stores NFC card and tag configuration details	config_id, family, card_instance, uid_type, uid_writable, auth_type, default_key_present, lock_state, ndef_protection, cve_flag, otp_set
attacks.csv	Stores attacker-tiered laboratory trial results	config_id, tier, repeat_no, read_success, write_success, emulation_success, compromised, time_to_compromise, censored, attempts, tool_version, timestamp, notes

Table 8: Dataset Structure for configs.csv and attacks.csv

3.4 Cox Regression-Based Weight Calibration

The suggested Exploitability-Calibrated NFC Security Profile score Model derives empirical score weights from laboratory-measured time-to-compromise data using Cox proportional hazards regression. This approach is employed because the study generates time-to-event data, with the successful breach of an NFC card or tag configuration serving as the event.

Every trial in this study documents whether the card was compromised and how long it took to do so. It's possible that some card combinations won't be compromised within the 1800-second testing window. Because these trials demonstrate that the card resisted compromise during the observation period, they are documented as suppressed observations. Because it can analyze both compromised and censored trials, Cox regression is appropriate.

The association between NFC security factors and compromise speed is examined using the Cox regression model. The likelihood of a quicker compromise may be increased by elements like writable UID, no authentication, the existence of a default key, unlocked memory, writable NDEF data, and known vulnerability indicators. A factor is given a higher coefficient in the model if it significantly helps to a quicker compromise.

For each attacker capability tier T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 the regression method is implemented independently. This is crucial because, depending on the attacker's tool capabilities, the same NFC security factor may have varied effects. For the purpose of calculating the final risk score, the resulting coefficients are further normalized and transformed into scoring weights.

3.5 Risk Score Calculation and Classification

The final NFC security risk score is determined by the proposed Exploitability-Calibrated NFC Security Profile Scoring Model, which uses Cox proportional hazards regression to determine factor weights. This step's goal is to translate calibrated weights and NFC card security factors into a numerical value that researchers, auditors, and security administrators can comprehend.

Each attacker capability tier - T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 has its own score determined by the scoring algorithm. This is significant since the same NFC card or tag may have varying risk levels based on the attacker's accessible tool. A card might be medium or high risk versus Flipper Zero or Proxmark3, but low risk against a smartphone.

The weighted total of the chosen NFC security parameters is used to determine the risk score. Each factor value is multiplied by the weight determined using Cox regression. Next, a 0–100 score is created by normalizing the entire weighted value.

$$\text{Risk Score} = (\sum \text{Weight} \times \text{Factor Value} / \sum \text{Maximum Possible Weighted Value}) \times 100$$

The coded security status of the NFC card or tag is represented by the factor value in this calculation. The codes for safe, medium-risk, and risky conditions are 0, 0.5, and 1, respectively. Based on measured time-to-compromise statistics, the weight indicates how important that factor is.

Three risk levels are assigned to the computed score. Low risk is defined as a score between 0 and 33, medium risk as a score between 34 and 66, and high risk as a score between 67 and 100. Users can readily comprehend the security posture of the tested NFC card or tag thanks to this classification.

The card family, attacker tier, computed risk score, risk classification, and a brief description of the primary elements influencing the outcome are all included in the final output. As a result, the scoring methodology is comprehensible and practical for evaluating NFC/RFID security.

Attacker-Tiered Risk Scoring for the Same NFC Card Configuration			
NFC Card Configuration	T1 – Smartphone	T2 – Flipper Zero	T3 – Proxmark3
NTAG215	Basic NFC App (e.g., NFC Tools)	Multi-protocol tool (e.g., Read / Write / Emulate)	Advanced research tool (e.g., Full memory access)
Card Family : NTAG215 UID Writable : Yes Authentication : None Default Key : No Memory Lock : Unlocked NDEF Protection : None Known Vulnerability : No	Risk Score (0–100) 25	Risk Score (0–100) 58	Risk Score (0–100) 82
	Risk Level Low	Risk Level Medium	Risk Level High
	Lower Risk Increasing Attacker Capability Higher Risk		

Figure 3: Illustrative Tier Matrix

Figure 3 illustrates how various attacker capability tiers can result in varying risk rankings for the same NFC card or tag. Instead of providing a single general risk score, this attacker-tiered view offers a more accurate security evaluation.

3.6 Validation Strategy and Performance Metrics

To ensure that the computed risk score accurately represents the exploitability of NFC cards and tags, the suggested Exploitability-Calibrated NFC Security Profile Scoring Model needs to be verified. During laboratory testing, the validation procedure determines whether lower-risk cards withstand compromise for a longer period of time and whether higher-risk cards are compromised more quickly.

The suggested score will be contrasted with baseline techniques including expert rating, CVSS-adapted scoring, and an unweighted checklist. The relationship between the determined risk score and the time-to-compromise is examined using Spearman correlation. The model's ability

to function with unseen NFC card families is tested using leave-one-family-out validation. To determine whether minor adjustments to weights or criteria have an impact on the final Low, Medium, or High categorization, sensitivity analysis is utilized.

Testing the same NFC card configuration several times is another method of reproducibility testing. The system should generate the same or very consistent risk score if the same input data is used. Lastly, the suggested scoring model is tested to see if it outperforms baseline techniques and offers a useful attacker-tiered NFC security evaluation.

Table 9 presents the main evaluation metrics used to validate the proposed

Evaluation Metric	Purpose	Target Outcome
Spearman correlation	Measures the relationship between risk score and measured exploitability	Higher-risk scores should match faster compromise results
Leave-one-family-out validation	Tests whether the model generalizes to unseen NFC card families	Model should remain stable when one card family is excluded
Sensitivity analysis	Tests whether score bands remain stable when weights or thresholds change	Low/Medium/High classification should not change significantly
Reproducibility testing	Tests whether repeated testing gives consistent results	Same card configuration should produce consistent or identical score
Baseline comparison	Compares proposed model with checklist, CVSS-adapted, and expert rating methods	Proposed model should better reflect measured time-to-compromise

Table 9: Evaluation Metrics and Target Outcomes

Validation Test	Description	Expected Result
Baseline comparison	Compare proposed score with unweighted checklist, CVSS-adapted score, and expert rating	Proposed score should better reflect measured compromise difficulty
Spearman rank test	Compare score ranking with time-to-compromise ranking	Higher-risk cards should generally be compromised faster
Cross-validation	Remove one card family and test model performance	Model should remain useful for unseen NFC families
Sensitivity test	Adjust factor weights and score thresholds	Classification should remain stable
Repeated scan test	Test same card configuration multiple times	Same input should give same or highly consistent output

Table 10: summarizes the Validation Tests for the Proposed Scoring Model

3.7 Tools and Platforms with Justification

To gather NFC card security data, compile datasets, determine statistical weights, compute risk scores, and present findings, the suggested Exploitability-Calibrated NFC Security Profile Scoring Model needs both hardware and software tools. The technologies that were chosen facilitate data analysis, attacker-tiered evaluation, controlled NFC/RFID laboratory testing, and reproducible prototype development.

A smartphone with NFC, Flipper Zero, and Proxmark3 are among the hardware tools. For basic writable tag testing, UID inspection, and NFC reading, the smartphone serves as the attacker capability tier. The practical attacker capability tier for field-level validation and authorized NFC observation is Flipper Zero. Because Proxmark3 enables controlled laboratory security testing and deeper NFC/RFID protocol-level analysis, it is utilized as the advanced attacker capability tier.

Python, pandas, NumPy, lifelines, Matplotlib, CSV files, and a development environment like Visual Studio Code or Jupyter Notebook are among the software tools. The scoring prototype is constructed with Python. Configs.csv and assaults are managed using pandas and CSV files.csv. Matplotlib is used to generate charts and visual results, NumPy facilitates numerical computations, and Lifelines facilitates Cox proportional hazards regression.

Tool / Technology	Purpose	Justification
Smartphone with NFC	Basic NFC reading and simple tag testing	Represents the basic attacker capability tier
Flipper Zero	Practical NFC testing and supported validation	Represents the practical attacker capability tier
Proxmark3	Advanced NFC/RFID protocol-level analysis	Represents the advanced attacker capability tier
Python	Prototype development and scoring calculation	Supports reproducible data analysis and implementation
pandas	CSV data loading, cleaning, and merging	Useful for managing configs.csv and attacks.csv
NumPy	Numerical calculations	Supports score normalisation and weighted calculations
Matplotlib	Charts and result visualisation	Used to present scoring and validation results
CSV files	Dataset storage	Simple, portable, and easy to inspect
VS Code	Development and analysis environment	Supports coding, testing, and demonstration

Table 11: Tools and Technologies Used in the Scoring Model

3.8 Ethical Considerations and Data Collection Limitations

NFC/RFID security testing is part of the suggested Exploitability-Calibrated NFC Security Profile Scoring Model. As a result, throughout the research, ethical and legal issues are crucial. Only authorized laboratory settings will be used for all testing, and the study team's NFC cards and tags will be used.

Real student ID cards, staff access cards, bank cards, transport cards, hotel key cards, office access cards, and any third-party NFC credentials will not be used in this study. This restriction is put in place to safeguard confidentiality, stop illegal access, and guaranty that the study stays within the bounds of defensive cybersecurity.

Only controlled testing and attacker-tier representation will make use of the research's tools, which include smartphone NFC apps, Flipper Zero, and Proxmark3. They won't be utilized for unapproved testing of real-world systems, unauthorized cloning, credential theft, or unauthorized access.

Only technical experimental values, including card family, configuration state, attacker tier, read success, write success, compromised status, time-to-compromise, censored status, number of attempts, tool version, and testing comments, will be included in the gathered dataset. Sensitive organizational data, real access credentials, financial information, or personal information will not be included in the dataset.

The experiment's use of specific NFC card families and readily available lab test cards is one of the study's limitations. As a result, not all NFC/RFID cards used in real-world settings may be represented by the results. The model's usage of three specific attacker capability tiers—smartphone, Flipper Zero, and Proxmark3—is another drawback. Although these tiers offer a useful comparison, they do not account for all potential attacker tools or situations.

Notwithstanding these drawbacks, the suggested approach offers a controlled, repeatable, and moral means of assessing NFC card security exposure. The study upholds both technical validity and ethical responsibility through the use of authorized test cards, organized datasets, attacker-tiered trials, and time-to-compromise metrics.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed research introduces an NFC Card and Tag Security Enhancement Framework Using Hybrid Detection that combines four independent but complementary components: C1 - NFC Security Profile Scoring, C2 - Multi-Session Protocol-Response Provenance Fingerprinting, C3 - NFC Tag Data Integrity Verification, and C4 - Hybrid Behavioral Anomaly Detection. Each component addresses a different NFC security problem while contributing to a unified assessment.

The framework begins with authorized NFC cards and tags, with data collected through a shared acquisition layer using devices such as Proxmark3 RDV2, Flipper Zero, and ACR122U. After preprocessing, the collected evidence is provided to the four components in parallel, where each component performs its analysis independently and produces a component-specific result.

Finally, the system stores the outputs in a shared backend and presents them through a unified dashboard and reporting layer. This architecture maintains component independence while providing an integrated view of NFC card and tag security.

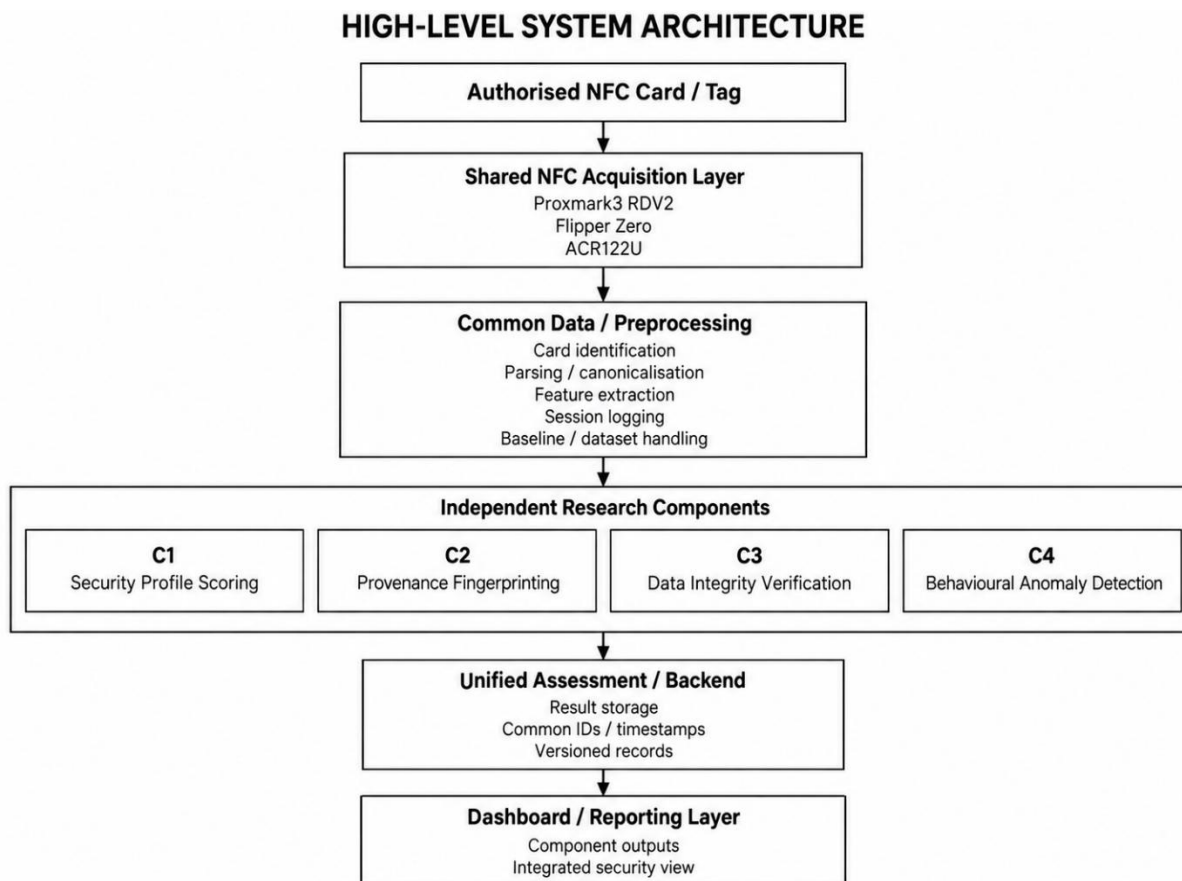


Figure 4: High-level architecture of the four-component framework

4.2 Internal Architecture of the Exploitability-Calibrated NFC Security Profile Scoring Model

The suggested NFC/RFID Card and Tag Security Enhancement Framework's NFC Security Profile Scoring component is the Exploitability-Calibrated NFC Security Profile Scoring Model component. This component's primary goal is to provide an attacker-tiered risk score and assess the security posture of NFC cards and tags. Card family, UID type, UID writability, authentication type, default key presence, memory lock state, NDEF protection status, and known vulnerability indications are among the NFC card and tag configuration data that the component takes as input.

Additionally, the Exploitability-Calibrated NFC Security Profile Scoring Model component makes use of laboratory experiment data gathered under three attacker capability tiers: T1 smartphone, T2 Flipper Zero, and T3 Proxmark3. The experiment logs read success, write success, emulation success, compromised status, time-to-compromise, censored status, and number of attempts for each attacker tier. Structured CSV datasets are used to hold the gathered configuration data and attacker-tiered trial data. Card configuration information is stored in the configs.csv file, and laboratory test results are stored in the attacks.csv file. The config_id variable is used to connect these two datasets.

Cox proportional hazards regression is used to extract empirical factor weights from time-to-compromise data once the dataset has been prepared. The scoring system then uses the calculated weights to determine each attacker tier's risk score, which ranges from 0 to 100. A tier-wise NFC security risk result, comprising the attacker tier, computed score, Low/Medium/High risk classification, and a brief description of the primary reasons influencing the score, is the final output of the Exploitability-Calibrated NFC Security Profile Scoring Model .

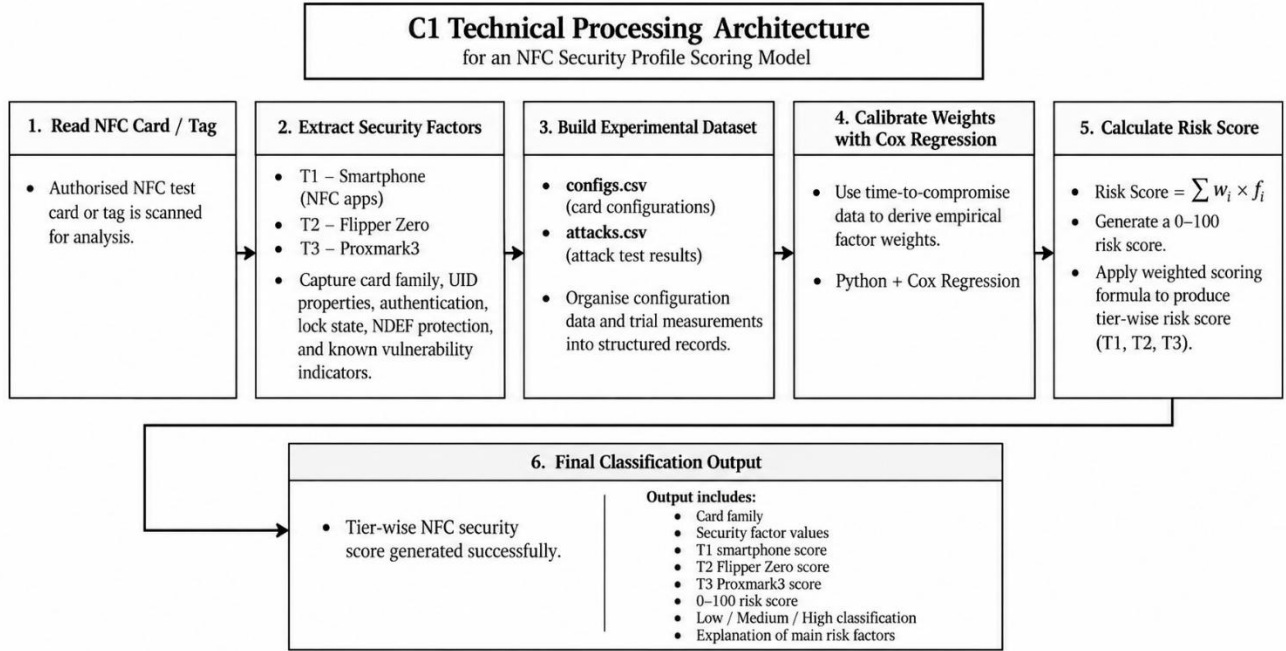


Figure 5: Internal Architecture of the Exploitability-Calibrated NFC Security Profile Scoring Model

Figure 5 depicts the internal workflow of the Exploitability-Calibrated NFC Security Profile Scoring Model. Authorized test cards are used to gather NFC card and tag security elements, laboratory trial results are recorded under T1, T2, and T3 attacker tiers, and the obtained data is processed using Cox regression to produce calibrated weights. Lastly, the scoring system determines the 0 - 100 risk score and assigns each attacker tier a Low, Medium, or High rating.

4.3 Data Flow of the System

The authorized NFC card and tag inputs start the data flow of the proposed Exploitability-Calibrated NFC Security Profile Scoring Model . Three attacker capability tiers T1 smartphone, T2 Flipper Zero, and T3 Proxmark3—are used to configure and test certain cards, including NTAG, MIFARE Ultralight, MIFARE Classic, and MIFARE DESFire. Security-related outcomes, including read success, write success, emulation success, compromised status, time-to-compromise, censored status, and number of attempts, are recorded by the system during each trial.

Two structured CSV datasets contain the gathered data. NFC card configuration information, including card family, UID type, UID writability, authentication type, default key present, lock status, NDEF protection, and known vulnerability indicators, is stored in the configs.csv file. Each attacker tier's laboratory testing findings are stored in

the attacks.csv file. The config_id column is used to join these two datasets before they are ready for Cox proportional hazards regression.

Following data preparation, the Cox regression model examines the correlation between recorded time-to-compromise values and NFC security criteria. The scoring engine receives the calculated weights and computes a 0–100 risk score for each attacker tier independently. The card family, attacker tier, computed score, Low/Medium/High risk rating, and the primary criteria influencing the score are all included in the system's final output report.

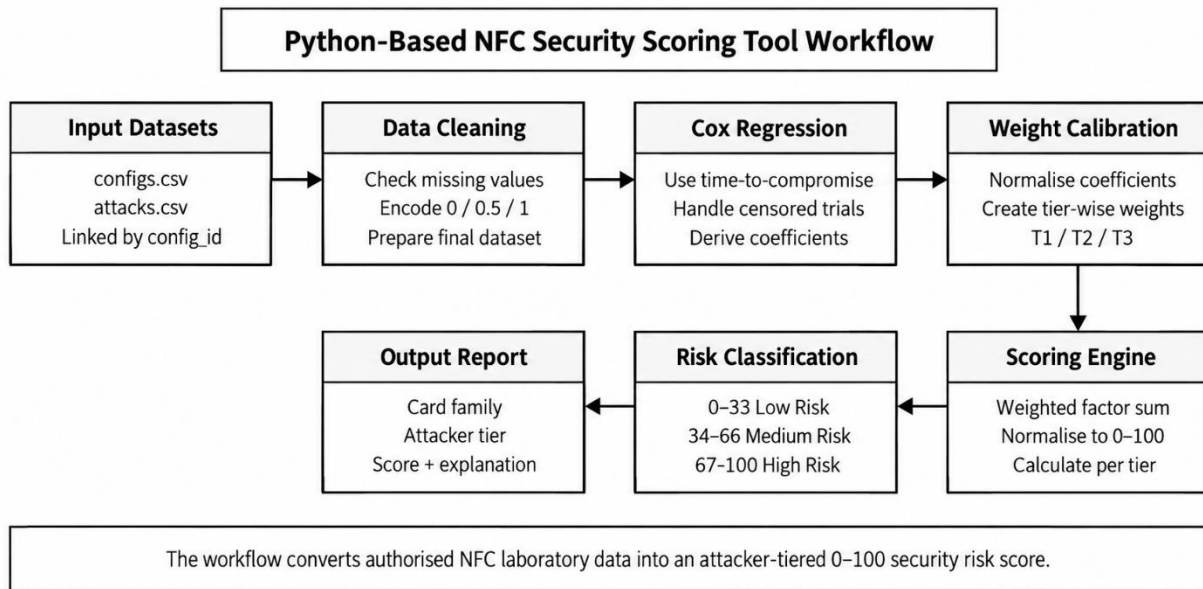


Figure 6: Python-Based NFC Security Scoring Tool Workflow

The Python-based NFC security scoring tool's data flow is depicted in Figure 6. It describes the loading, cleaning, merging, Cox regression analysis, and conversion of configs.csv and attacks.csv into an attacker-tiered 0–100 risk score.

4.4 Scalability Considerations

The proposed Exploitability-Calibrated NFC Security Profile Scoring Model component is designed to be scalable, enabling future growth without changing the overall scoring model. The current study takes into account NFC card families as NTAG, MIFARE Ultralight, MIFARE Classic, and MIFARE DESFire.

However, more NFC card families can be added later by adding more records to the configs.csv table.

Additional security criteria and attacker capability tiers can be added to the model. UID writability, authentication type, default key existence, memory lock state, NDEF protection, and known vulnerability indications are among the variables used in the current model. additional time-to-compromise data can be used to recalibrate the Cox regression model and add additional NFC security features or testing tools to the dataset.

Python and CSV-based datasets are used in the suggested solution, which facilitates system modification and enhancement. The prototype may eventually be expanded into a desktop program, online application, or API-based NFC security evaluation tool. Because it supports additional cards, new factors, new attacker tiers, and larger datasets, The Exploitability-Calibrated NFC Security Profile Scoring Model is scalable.

4.5 Security Considerations

Only defensive NFC/RFID security assessments in approved laboratory settings are intended to use the proposed Exploitability-Calibrated NFC Security Profile Scoring Model. The study team's NFC cards and tags are the only ones used for testing. There will be no testing of actual student ID cards, staff access cards, bank cards, transport cards, hotel key cards, workplace access cards, or third-party credentials.

Only technical research data, including card family, configuration state, attacker tier, read and write success, compromised status, time-to-compromise, and censored status, will be stored in the gathered dataset. Sensitive organizational data, financial information, actual access credentials, and personal information will not be stored. This lowers security risks and protects privacy.

The research's tools, such as Flipper Zero, Proxmark3, and smartphone NFC apps, are only utilized for attacker-tier representation and controlled testing. Risk scoring, security posture evaluation, validation findings, and defensive suggestions will be the main topics of the final report. It won't offer guidance on how to utilize NFC/RFID systems improperly, steal credentials, or perform illegal cloning.

5. USER REQUIREMENTS

5.1 Functional Requirements

The primary tasks that the proposed Exploitability-Calibrated NFC Security Profile Scoring Model security scoring tool must accomplish are outlined in the functional requirements. The system must allow data input, data processing, score computation, attacker-tiered classification, and result reporting because this component focuses on NFC card and tag security scoring.

The researcher must be able to import NFC card configuration data from the configs.csv dataset into the suggested system. Card family, UID type, UID writability, authentication type, default key presence, memory lock state, NDEF protection status, known vulnerability indications, and OTP status are all included in this data. Attacker tier, read success, write success, emulation success, compromised status, time-to-compromise, censored value, number of attempts, tool version, timestamp, and comments must all be imported by the system from the attacks.csv dataset.

Prior to calculating the score, the system must verify and clean the data after importing the datasets. It must compute a 0–100 risk score independently for the T1 smartphone, T2 Flipper Zero, and T3 Proxmark3 attacker tiers, encode NFC security parameters into numerical values, and use Cox proportional hazards regression to determine empirical weights. After that, the computed score needs to be categorized as Low Risk, Medium Risk, or High Risk.

A clear scoring report must be the system's end product. The card family, attacker tier, calculated risk score, risk classification, and a brief summary of the key elements that affected the outcome should all be included in the report. Additionally, the results should be able to be exported or saved for later analysis and documentation.

Table 12 summarises the main functional requirement of the proposed Exploitability Calibrated NFC Security profile Scoring Model. These specifications outline the fundamental functions that the system must carry out in order to determine and report the security risk associated with NFC cards.

Requirement ID	Functional Requirement
FR1	The system shall import NFC card configuration data from configs.csv.
FR2	The system shall import attacker-tiered laboratory test results from attacks.csv.
FR3	The system shall validate and clean the imported dataset.
FR4	The system shall encode NFC security factors into numerical values.
FR5	The system shall apply Cox regression to derive scoring weights.
FR6	The system shall calculate a 0–100 risk score for each attacker tier.
FR7	The system shall classify the score as Low Risk, Medium Risk, or High Risk.
FR8	The system shall generate an output report with score explanation.
FR9	The system shall support saving or exporting the scoring results.

Table 12: Functional Requirements of the Proposed Scoring Tool

5.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes of the proposed Exploitability-Calibrated NFC Security Profile Scoring Model . Rather than just outlining what the system does, these criteria specify how well it should function.

The suggested grading system needs to be precise, dependable, and repeatable. The measured exploitability of NFC cards and tags should be reflected in the computed risk score. The system should yield the same or very similar results when the same NFC card setup and input dataset are utilized. This is crucial because repeatable security assessment must be supported by the model.

Additionally, the system needs to be explainable, safe, maintainable, and usable. Researchers and security managers should have little trouble understanding the tool. Sensitive organizational information, financial data, genuine credentials, and personal information should not be stored there. Future additions of new NFC card varieties, security features, and attacker levels should also be supported by the system. The final product should provide an explanation for the card's classification and risk score.

Table 13 summarises the non-functional requirements of the proposed Exploitability -Calibrated NFC Security Profile Scoring Model. Accuracy, dependability, usability, security, maintainability, efficiency, and explainability are all aspects of system quality that are the emphasis of these standards.

Requirement ID	Non-Functional Requirement
NFR1	The system shall produce accurate risk scores based on calibrated factor weights.
NFR2	The system shall generate consistent results for the same input dataset.
NFR3	The system shall be easy to use and understand for researchers and security administrators.
NFR4	The system shall not store personal data, real credentials, or sensitive organisational data.
NFR5	The system shall support future extension with new card families, factors, and attacker tiers.
NFR6	The system shall process CSV-based datasets within a reasonable time.
NFR7	The system shall provide explainable score outputs and risk classifications.

Table 13: Non-Functional Requirements of the Proposed Scoring Tool

6. COMMERCIALIZATION PLAN

As a defensive NFC/RFID security assessment tool, the suggested Exploitability-Calibrated NFC Security Profile Scoring Model has commercialization potential. NFC and RFID cards are used by many organizations for asset tracking, hotel access, transportation, attendance, and access control. These companies might not, however, have an easy way to determine if their NFC cards are fully, partially, or weakly protected.

The suggested novelty component's primary benefit is its ability to translate technical NFC card setup data into a straightforward risk assessment between 0 and 100. Administrators, auditors, and non-technical decision makers can easily comprehend this score. For attacker tiers such as smartphones, Flipper Zero, and Proxmark3, the tool offers a Low Risk, Medium Risk, or High Risk assessment independently rather than just displaying raw NFC card details.

Universities, businesses, hotels, cybersecurity training facilities, and organizations that use NFC-based cards or smart tags can all benefit from the suggested tool's commercialization as a security assessment service. A report detailing the card family, risk score, risk level, and suggested security enhancements can be produced by an authorized security auditor who tests sample cards from an organization.

A desktop program, web application, or API-based NFC security evaluation platform could be developed from the Python-based prototype in the future. Additionally, clone detection, data integrity verification, and behavioral anomaly detection can be supported by integrating the tool with the broader NFC/RFID Card and Tag Security Enhancement Framework.

.

7. BUDGET AND JUSTIFICATION

The C2 budget is based on the shared hardware and software resources available within the overall NFC Card and Tag Security Enhancement Framework. Although these resources are shared across the wider research project, this section explains their specific relevance to C2 - Multi-Session Protocol-Response Provenance Fingerprinting.

C2 requires repeatable NFC interaction data collected across multiple controlled sessions. The main costs therefore relate to NFC acquisition hardware, authorized test cards, and supporting analysis tools. Most software used for data processing, feature extraction, statistical analysis, and machine-learning evaluation is free or open source.

Item	Category	Estimated Cost (LKR)	C2-Specific Justification
Proxmark3	Research Hardware	25,000	Primary reader for repeated protocol-response and timing acquisition.
Flipper Zero	Research Hardware	80,000	Authorized emulation source for genuine-vs-emulated comparison.
NFC Card Set	Test Media / Consumables	10,000	Multiple authorized card families for repeatable C2 experiments.
Python 3.x	Software	Free	Data processing, feature extraction, and model development.
Pandas / NumPy	Software	Free	Cleans and summarizes repeated session measurements.
scikit-learn	Software	Free	Implements and evaluates RF, SVM, and KNN models.
Jupyter Notebook	Software	Free	Exploratory analysis and model evaluation.
Proxmark3 Client / Iceman Firmware	Software	Free	Controls Proxmark3 and supports NFC data acquisition.
Git / GitHub	Development Platform	Free	Version control and experiment tracking.
Human Effort	Research Effort	Not monetized	Student-led data collection, analysis, and documentation.

Table 14: Budget and justification.

8. WORK BREAKDOWN STRUCTURE (WBS)

The suggested Exploitability-Calibrated NFC Security Profile Scoring Model is broken down into doable research activities according to the Work Breakdown Structure. Background research, methodology design, laboratory testing, dataset preparation, Cox regression analysis, scoring tool development, validation, and documentation are all included in the project.

The literature review and identification of research gaps are the main objectives of the first stage. In this phase, attacker capacity levels, general scoring systems, NFC analysis tools, and current NFC/RFID security vulnerabilities are examined. This supports the necessity for an NFC security score model that is data-driven.

The design of the approach and laboratory setup are the main topics of the second stage. Choosing NFC card families, defining security configuration states, determining attacker capability levels, creating authorized test cards, and selecting the metrics needed for time-to-compromise analysis are all part of this process.

Data gathering and dataset preparation are the main topics of the third stage. Smartphones, Flipper Zero, and Proxmark3 attacker tiers are used to test authorized NFC test cards. The gathered data is cleaned, encoded, and ready for Cox proportional hazards regression before being saved in `configs.csv` and `attacks.csv`.

Model development and validation are the main topics of the fourth stage. Empirical factor weights are obtained from measured time-to-compromise data using Cox regression. The 0–100 risk score is then determined using a Python-based scoring prototype, which then categorizes each outcome as Low Risk, Medium Risk, or High Risk. Sensitivity analysis, repeatability testing, baseline comparison, and Spearman correlation are used to validate the final model.

Documentation and submission are the main topics of the last phase. This entails creating the proposal, diagrams, tables, final dissertation content, evaluation findings, appendices, references, proofreading, and final formatting.

Table 15 summarises the main tasks included in the Work Breakdown Structure of the proposed Exploitability-Calibrated NFC Security Profile Scoring Model.

WBS ID	Task	Description
WBS 1	Literature Review	Study NFC/RFID security issues, existing tools, and scoring frameworks
WBS 2	Research Gap Identification	Identify limitations in current NFC card security scoring approaches
WBS 3	Methodology Design	Define card families, security states, attacker tiers, and measurements
WBS 4	Laboratory Setup	Prepare authorised NFC test cards, devices, and testing environment
WBS 5	Data Collection	Record attacker-tiered trial results and time-to-compromise values
WBS 6	Dataset Preparation	Clean, encode, and structure configs.csv and attacks.csv
WBS 7	Cox Regression Analysis	Derive empirical factor weights from time-to-compromise data
WBS 8	Scoring Tool Development	Build Python prototype for 0–100 risk score calculation
WBS 9	Validation and Evaluation	Compare proposed model with baseline methods and test reproducibility
WBS 10	Documentation and Submission	Prepare proposal, dissertation, diagrams, tables, and final report

Table 15: Work Breakdown Structure

Figure 8 visually represents the Work Breakdown Structure of the proposed Exploitability-Calibrated NFC Security Profile Scoring Model . It shows the main project tasks in a hierarchical format.

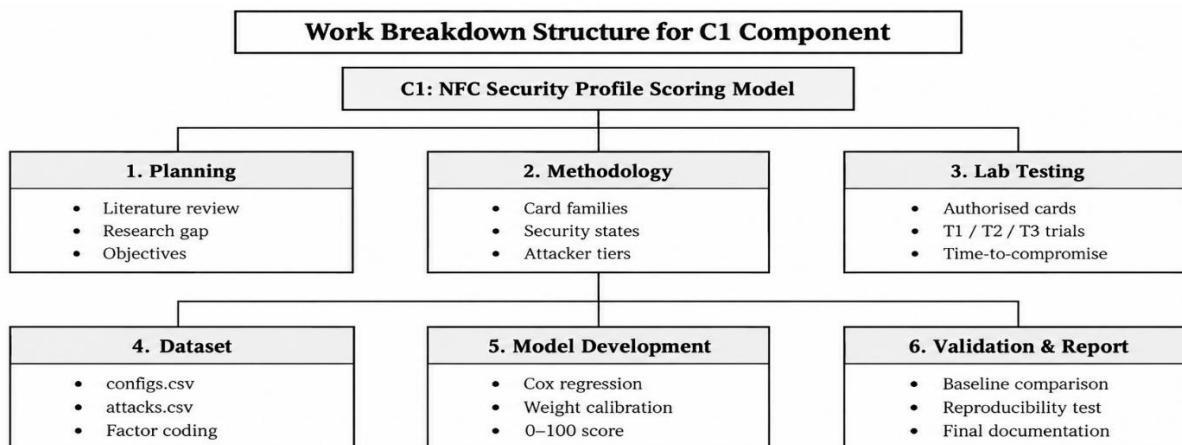


Figure 7: Work Breakdown Structure

9. GANTT CHART

9.1 Project Timeline

The planned timetable for finishing the suggested Exploitability-Calibrated NFC Security Profile Scoring Model is described in the project timeline. A defined timeframe is necessary to effectively organize the task because this research involves a literature evaluation, methodology design, laboratory testing, dataset preparation, statistical analysis, prototype construction, validation, and documentation.

The literature review and identification of research gaps are the main objectives of the first stage. In this phase, attacker capability levels, NFC analysis tools, security score methods, and current NFC/RFID security concerns are examined. This aids in defining the suggested scoring model's uniqueness.

The design of the approach and laboratory setup are the main topics of the second stage. Choosing NFC card families, specifying security configuration states, creating authorized NFC test cards, determining attacker tiers, and choosing the metrics needed for time-to-compromise analysis are all part of this process.

The production of datasets and laboratory testing are the main topics of the third stage. Smartphone, Flipper Zero, and Proxmark3 attacker tiers are used to test authorized NFC test cards. After being cleaned, encoded, and ready for analysis, the gathered data is saved as `configs.csv` and `attacks.csv`.

Model development and validation are the main topics of the fourth stage. Empirical factor weights are obtained from time-to-compromise data using Cox proportional hazards regression. The 0–100 risk score is then determined by the Python-based scoring prototype, which then categorizes each outcome as Low Risk, Medium Risk, or High Risk. Sensitivity analysis, repeatability testing, baseline comparison, and Spearman correlation are used to validate the model.

Documentation and final submission are the main topics of the last phase. The proposal, tables, diagrams, references, appendices, final report content, proofreading, formatting, and submission are all included in this.

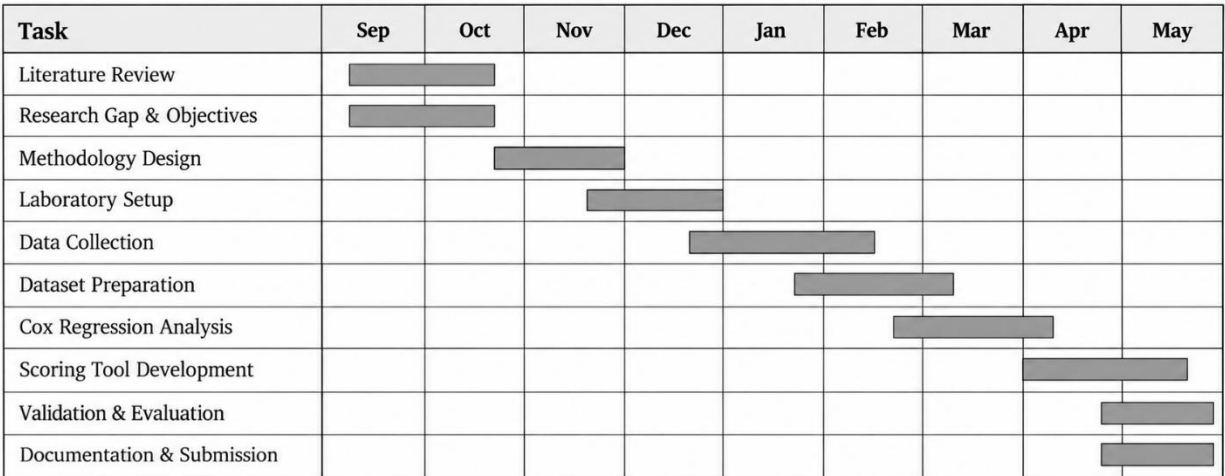


Figure 8:Gantt Chart

Figure 8 displays the suggested scoring model's projected timeline. From literature study and methodology design to laboratory testing, model building, validation, documentation, and final submission, it outlines the primary project tasks.

9.2 Risk Register and Contingency Plan

Risk ID	Possible Risk	Impact	Mitigation / Contingency Plan
R1	NFC testing device failure	Delay in data collection	Check smartphone, Flipper Zero, and Proxmark3 before testing and allocate troubleshooting time
R2	Limited card configuration support	Some test states may not be available	Mark unavailable configurations as not applicable in the dataset
R3	Missing or inconsistent dataset values	Incorrect score calculation	Apply data cleaning and validation before analysis
R4	Small or unbalanced dataset	Weak statistical calibration	Conduct repeated trials where possible and compare with baseline methods
R5	Misuse of NFC/RFID testing tools	Ethical and legal issues	Use only authorised laboratory test cards and avoid real credentials

Table 16: Risk Register and Contingency Plan

The risk registers aids in the responsible and controlled management of the study. Early detection of technological, ethical, data-related, and time-management risks allows the project to be finished more efficiently while preserving the quality and safety of the research.

10. REFERENCES

- [1] G. H. B. Damasceno, I. A. S. Tacca, G. P. Aquino, and E. C. Vilas Boas, “Cloning and Tampering with MIFARE Classic Cards: An Empirical Assessment in Live Systems,” in *Proc. 2025 5th International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)*, 2025, pp. 527–532, doi: 10.1109/ICICYTA68677.2025.11362828.
- [2] R. Habrah, K. Hassan, A. Almallah, F. Alkaisi, and F. Kharbat, “A Systematic Literature Review on Cyber-Physical Security Vulnerabilities in RFID-based Access Control Systems,” in *Proc. 2025 12th International Conference on Social Networks Analysis, Management and Security (SNAMS)*, 2025, pp. 281–288, doi: 10.1109/SNAMS67467.2025.11391156.
- [3] T. V. Morozova and V. V. Gurov, “Research in RFID Vulnerability,” in *Proc. IEEE Conference*, 2017, pp. 501–503.
- [4] P. Zlatarov and G. Ivanova, “Design and Development of a Smartphone-enabled Smart Card-based Attendance Tracking Module for Personalized Education,” in *Proc. 2023 7th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, 2023, doi: 10.1109/ISMSIT58785.2023.10304892.
- [5] R. M. N. D. Ranasinghe and G. Z. Yu, “RFID/NFC Device with Embedded Fingerprint Authentication System,” in *Proc. IEEE Conference*, 2017, pp. 266–269.
- [6] J. Bravo, R. Hervás, G. Chavira, and S. Nava, “Adapting Technologies to Model Contexts: Two Approaches through RFID & NFC,” in *Proc. IEEE Conference*, 2007.
- [7] A. Luntovskyy, T. Zobjack, B. Shubyn, and M. Klymash, “Energy Efficiency and Security for IoT Scenarios via WSN, RFID and NFC,” in *Proc. 2021 IEEE International Conference on Information and Telecommunication Technologies and Radio Electronics (UkrMiCo)*, 2021, doi: 10.1109/UkrMiCo52950.2021.9716591.
- [8] P. Dutta, J. Zhang, B. Amoah, X. Wang, K. K. Podder, and S. Mao, “Enhancing RFID Observations Using Masked Autoencoders for Improved Indoor Localization,” in *Proc. 2024 IEEE International Conference on RFID Technology and Applications (RFID-TA)*, 2024, pp. 149–152, doi: 10.1109/RFID-TA64374.2024.10965153.

- [9] B. Amoah, X. Wang, J. Zhang, S. Mao, S. C. G. Periaswamy, and J. Patton, “Optimizing Adaptive Power Control for Enhancing Robustness in RFID Sensing,” in *Proc. 2024 IEEE International Conference on RFID Technology and Applications (RFID-TA)*, 2024, pp. 157–160, doi: 10.1109/RFID-TA64374.2024.10965152.
- [10] J. M. Faria, L. Vilas Boas, J. Dillen, N. Simoes, I. Caetano, L. Cardoso, J. Borges, and A. H. J. Moreira, “RFID Tag Motion Detection through Machine Learning with a Single Antenna Setup,” in *Proc. 2024 IEEE International Conference on RFID Technology and Applications (RFID-TA)*, 2024, pp. 66–69, doi: 10.1109/RFID-TA64374.2024.10965163.
- [11] J. Bhanushali, T. Khan, K. L. Sudha, R. Singh, and C. Devadiga, “Tagd India: NFC Enabled Smart Visiting Card,” in *Proc. 2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)*, 2024, doi: 10.1109/ISCS61804.2024.10581315.
- [12] J. Sun, A. Diet, Y. Le Bihan, and M. Police, “A 3-D RFID/NFC Cylindrical Magnetic Reader Coils Structure, 3DCCS, for Orientation-Free Detection of HF Tags, for Biomedical Applications,” *IEEE Journal of Radio Frequency Identification*, vol. 9, pp. 236–246, 2025, doi: 10.1109/JRFID.2025.3570909.
- [13] K. Kashyap and P. Pandey, “RFID Coupler Design for Encoding RFID Tags without Retraction inside Printers,” in *Proc. 2025 IEEE International Conference on RFID Technology and Applications (RFID-TA)*, 2025, doi: 10.1109/RFID-TA63091.2025.11265851.

11. APPENDICES

Appendix A - Dataset Schema and Field Definitions

Appendix A.1 - configs.csv

The configs.csv dataset stores the configuration and security properties of each authorized NFC card or tag used in the controlled laboratory experiment.

Fields:

config_id
family
card_instance
uid_type
uid_writable
auth_type
default_key_present
lock_state
ndef_protection
cve_flag
otp_set

Appendix A.2 - attacks.csv

The attacks.csv dataset stores the attacker-tiered laboratory trial results for each NFC card or tag configuration. The config_id field links the trial results with the corresponding configuration stored in configs.csv.

Fields:

config_id
tier
repeat_no
read_success
write_success
emulation_success
compromised
time_to_compromise
censored
attempts
tool_version

timestamp
notes

The two structured datasets provide the configuration factors and measured time-to-compromise information required for Cox proportional hazards regression and the final attacker-tiered NFC security risk score.

Appendix B - Validation and Evaluation Checklist

The following validation activities are used to evaluate the proposed NFC Security Profile Scoring Model:

Baseline Comparison

Compare the proposed scoring model with an unweighted checklist, CVSS-adapted scoring, and expert rating.

Spearman Rank Test

Evaluate the relationship between the calculated risk score and measured time-to-compromise.

Leave-One-Family-Out Validation

Exclude one NFC card family during model development and evaluate the model using the unseen family.

Sensitivity Analysis

Examine whether reasonable changes to factor weights or score thresholds significantly affect the final Low, Medium, or High risk classification.

Reproducibility Testing

Repeat the same card configuration tests to determine whether identical or highly consistent risk scores are produced.

Appendix C - Ethical Testing Boundary

All NFC/RFID experiments conducted as part of this research are limited to authorised laboratory environments and NFC cards or tags owned or approved for use by the research team.

The research does not involve testing real student ID cards, staff access cards, bank cards, transport cards, hotel key cards, workplace access cards, or third-party credentials.

Smartphone NFC applications, Flipper Zero, and Proxmark3 are used only for controlled laboratory testing and attacker-tier representation. They are not used for unauthorised cloning, credential theft, unauthorised access, or testing against third-party systems.

The collected datasets contain only technical experimental information such as card family, configuration state, attacker tier, test outcomes, time-to-compromise, censored status, number of attempts, tool version, and testing notes. Personal information, real access credentials, financial information, and sensitive organisational data are excluded from the dataset.

Appendix D — Prototype Dashboard

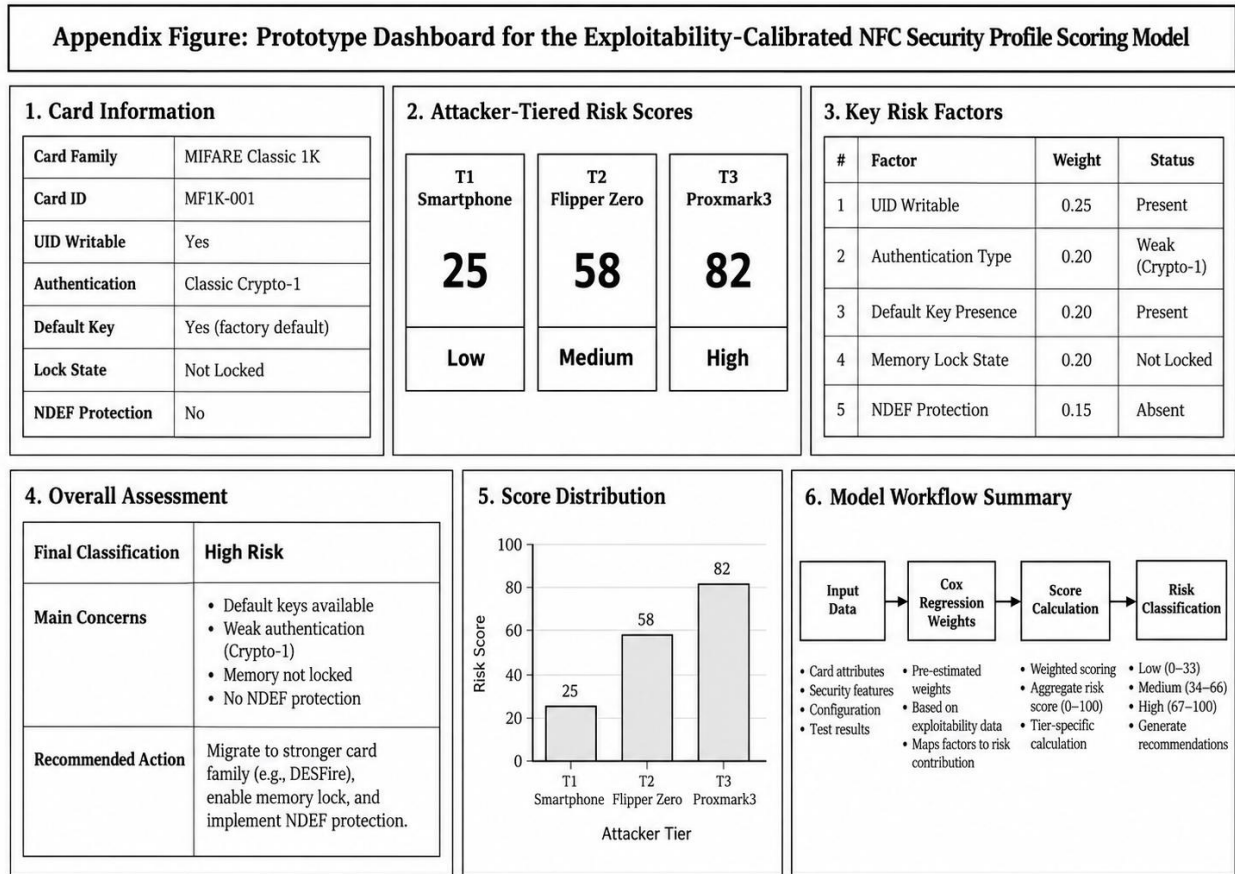


Figure 9 Prototype Dashboard

Appendix 2 - Mandatory Proposal Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
2026-08-20	Existing NFC/RFID tools mainly provide descriptive card information and do not produce a standard security risk score.	Literature review and comparison of smartphone NFC apps, Flipper Zero, Proxmark3, and existing security scoring approaches.	Use an existing generic scoring framework such as CVSS or an expert-defined checklist.	Develop an NFC-specific 0–100 Security Profile Scoring Model.	Meeting date : 2026-08-28.
2026-09-01	A single security score may not represent the different capabilities available to different attackers.	Review of smartphone NFC tools, Flipper Zero, and Proxmark3 capabilities.	Use one general risk score for every attacker.	Introduce three attacker capability tiers: T1 Smartphone, T2 Flipper Zero, and T3 Proxmark3.	Meeting date : 2026-08-28.
2026-09-10	Manually assigned scoring weights may be subjective and may create circular validation.	Review of risk-scoring approaches and proposed laboratory time-to-compromise measurements.	Expert-defined weights, equal weights, or CVSS-adapted weights.	Use Cox proportional hazards regression to derive empirical factor weights from measured time-to-compromise data.	Meeting date : 2026-09-02

Appendix 3. AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
GPT-5.6 Sol	Language and academic writing assistance	Grammar correction, sentence restructuring, clarity improvements, and academic wording for selected sections of the proposal.	AI-assisted text was reviewed against the original research concept, methodology, project documents, and cited academic sources before inclusion.	I reviewed, edited, accepted, or rejected all suggested wording and remains responsible for the final text.
GPT-5.6 Sol	Document structure and formatting assistance	Assistance with organizing sections, tables, appendices, abbreviations, figure captions, and presentation of technical content.	Suggested structures were compared with the official proposal template and the actual C1 research methodology.	I selected the final structure and ensured that it accurately represents the C1 component.
GPT-5.6 Sol	Technical explanation assistance	Assistance in explaining concepts such as attacker-tiered scoring, Cox proportional hazards regression, time-to-compromise, validation methods, and dataset structure.	Technical explanations were checked against the research methodology and the academic sources cited in the proposal.	I made the research decisions, selected the methodology, and ensured that the final explanation could be independently understood and defended.
GPT-5.6 Sol	Visual and diagram preparation assistance	Assistance in simplifying and preparing research workflow diagrams, architecture diagrams, scoring matrices, WBS, and project timeline visuals.	Each visual was checked against the written methodology, system architecture, datasets, and research workflow.	I determined the information included in each figure and verified that the diagrams represent the proposed C1 component correctly.